

## Class 10 · Practice Test — Paper T2

Time: 50 minutes · Maximum marks: 19 · Answer all questions. Show your working.

### Section A 1 mark

- Q1. Butanone is a four-carbon compound with the functional group [1]  
(a) carboxylic acid (b) aldehyde (c) ketone (d) alcohol
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### Section B 2 marks each

- Q2. Which term of the AP 21, 18, 15, ... is  $-81$ ? [2]
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- Q3. List two methods of producing magnetic fields. [2]
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- Q4. (a) What is meant by trophic level ? [2]  
(b) On an average, what percentage of energy is transferred to the second trophic level from the first trophic level ? Which categories of consumers do the organisms of the second trophic level belong to ?
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- Q5. Find the ratio in which the point  $P(-4, 6)$  divides the line segment joining the points  $A(-6, 10)$  and  $B(3, -8)$ . [2]
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### Section C 3 marks each

- Q5. The far point of a myopic person is 80 cm in front of the eye. What is the nature and power of the lens required to correct the problem? [3]
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- Q6. Pratyush took sulphur powder on a spatula and heated it. He collected the gas evolved by inverting a test tube over it. [3]  
(a) What will be the action of gas on (i) dry litmus paper? (ii) moist litmus paper?  
(b) Write a balanced chemical equation for the reaction taking place.
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### Section D case-based, 4 marks

- Q7. The taxi fare in a city consists of a fixed charge together with a charge proportional to the distance covered. For a distance of 10 km, the charge paid is ₹105 and for a journey of 15 km, the charge paid is ₹155. Let the fixed charge be ₹ $x$  and the per-km charge be ₹ $y$ . [4]  
(i) Form the pair of linear equations.  
(ii) Find the fixed charge and the per-km rate.  
(iii) How much does a person have to pay for travelling 25 km?
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